THE PHILOSOPHER'S ZETTELKASTEN

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The Philosopher's Zettelkasten

A Pocket Dictionary

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For all who aspire to learn.

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Keeping simplicity and clarity at the front of your mind is the only way to avoid the trap of your notes growing out of control.

— Will Simpson,
Zettelkasten Forum

INTRODUCTION

To the reader who has picked up this small book
and not yet put it down.
This is cause for modest celebration.

You hold in your hands a dictionary. Do not be alarmed. A dictionary is not a book one reads — or so the common wisdom insists. It is a book one *consults*, in the manner one consults a physician: briefly, reluctantly, and only when something has gone wrong. One does not, the argument proceeds, sit by the fire of an evening and read a dictionary from A to Z as though it were a novel with an exciting conclusion.

To this common wisdom I offer my respectful disagreement.

The dictionary is, in fact, the most honest of all literary forms. The novel dissembles. It pretends to have characters and a plot, when what it truly wishes to discuss is the human condition. The philosophical treatise erects magnificent scaffolding — premises, syllogisms, refutations, appendices — and buries its single useful insight somewhere in the middle, hoping the reader will not notice how little has been accomplished with how many pages. The sermon requires a congregation and a certain theatrical disposition. The dictionary alone says plainly: here is a word, here is what it means, here is why it matters. There is a bravery in alphabetical order that chronological argument rarely achieves.

This particular dictionary concerns itself with the **Zettelkasten** — a German word meaning, as we shall examine at proper length in its appointed place, a *box of slips*. That this humble object should require a

philosophical dictionary will strike some readers as extravagant. To those readers I say: you have clearly never lost an idea.

For what is the great tragedy of intellectual life? It is not ignorance. Ignorance is merely the beginning of the story, and stories beginning in ignorance have, historically, turned out rather well. No — the great tragedy is the *vanishing* of thought. The insight that arrives at three in the morning and is gone by breakfast. The connection between two ideas, glimpsed for a radiant moment across the crowded room of one's consciousness, that dissolves before one has found the words to introduce them to each other. The book read with passionate attention, annotated with furious marginalia, set upon the shelf — and forgotten within a season, leaving behind only the vague sensation that one has, somewhere, thought something important about something.

Memory is not a library. It is, if we are honest, something closer to a city after a moderate earthquake — much still standing, much collapsed, and considerable confusion about which was which. We remember what moved us, what frightened us, what embarrassed us at dinner parties. We remember rather less of what we deliberately studied, carefully read, and sincerely intended to retain. This is not a moral failing. It is an architectural one. The mind was not designed for the filing of abstractions. It was designed for the avoiding of large animals and the securing of dinner — tasks at which it remains, one must admit, impressively capable.

The Zettelkasten was invented — or rather, discovered, in the way that useful things are discovered, by someone sufficiently exasperated to look for them — as a remedy for precisely this condition. It is a system of **Index Cards**, each bearing a single thought, connected to other cards by threads of association, cross-reference, and the accumulated intelligence of a mind that has learned to externalize itself without embarrassment. It does not replace **Memory**. It civilizes it.

Now, a dictionary of this system might have been written as a manual. Many such manuals exist. They are written with great earnestness by people who genuinely wish to help, and they are read with great earnestness by people who genuinely wish to be helped, and the result is frequently a shelf of earnest manuals and a Zettelkasten that remains, as yet, unbegun.

This dictionary has been written instead as a work of **Philosophy** — by which I mean not the academic philosophy that produces papers read by eleven specialists and a graduate student who has wandered into the wrong seminar, but philosophy in its older and more companionable sense: the love of wisdom, practiced in the belief that wisdom is lovable, and that its pursuit need not require the abandonment of either **Clarity** or good humour.

Each entry in this dictionary takes a word from the vocabulary of the Zettelkasten — **Writing**, **Reading**, **Thinking**, **Learning**, **Systems**, **Language**, **Creativity**, and their assembled companions — and examines it as one might examine a stone picked up on a walk: turning it over, noticing its weight, considering what it is made of and how it came to be here and what would be lost if one simply put it down and walked on. Sometimes the examination yields a definition. Sometimes it yields a question. Sometimes it yields the unsettling suspicion that a word one has used with great confidence for many years means something rather different from what one supposed.

This last outcome is, I submit, among the most valuable a dictionary can produce.

A word must be said about **German Thoroughness** — that quality which gave us the Zettelkasten in its mature form, and which has shaped this dictionary's subject even where it has not shaped its style. The Germans have long understood something that other intellectual traditions approach only intermittently: that **Thinking** is a practice, not an inspiration. That the cultivation of ideas requires method as surely as the cultivation of a garden requires soil, water, and a willingness to get one's hands dirty. That **Creativity** is not a visitation from the muse — who is, frankly, unreliable — but a discipline, developed over time, supported by good tools and honest habits.

Niklas Luhmann, that most systematic of sociologists, maintained a Zettelkasten of ninety thousand cards and produced from it a body of work that would have occupied several ordinary lifetimes. He did not consider this remarkable. He considered it the natural consequence of working properly. We may find this either inspiring or exhausting, according to our temperament. In either case, we must concede the point.

And what of **Artificial Intelligence** — that great contemporary phenomenon which stores more than Luhmann, retrieves faster than any human hand, and generates text with a fluency that would have astonished Voltaire and perhaps alarmed him? Does the Zettelkasten survive in an age of machines that remember everything and forget nothing, that connect all ideas to all other ideas at velocities no index card can match?

It does. For reasons this dictionary will examine in their proper place, the question of *what* to think cannot be answered by any tool that is only concerned with *how quickly* to produce an answer. The machine is, in this respect, the ultimate flatterer: it tells you everything except what matters. The Zettelkasten asks the harder question. It asks you to decide.

This book is small. This is intentional.

Smallness is not a deficiency of ambition. It is a philosophical position. The great errors of intellectual history have generally been errors of scale — the system that explained everything, the theory that admitted no exceptions, the library so vast it could not be navigated, the treatise so long it could not be finished. Pascal's pensées, Montaigne's essays, the aphorisms of Marcus Aurelius — the works that have accompanied humanity longest have tended to fit, in some form or another, in a coat pocket.

The index card is small by design. The thought it carries should be able to stand on its own — which means it must be complete, and honest, and unpadded by the kind of elaborate qualification that signals not careful thinking but anxious thinking. To write on an index card is to commit. To fill a box with index cards is to accumulate commitments. To connect those cards into a system is to discover, sometimes with surprise and sometimes with dismay, what one actually believes.

This dictionary is small for the same reason. It does not intend to say everything. It intends to say something — clearly, and with enough **Clarity** that the reader may use it as a beginning rather than an ending. I ask of you only this: read what is useful, dispute what seems wrong, discard what does not serve you, and keep what does. This is, after all, precisely what a Zettelkasten is for.

Figure 1. Test the efficiency of your Zettelkasten by searching for specific information. If you can quickly locate what you need, your system is effective.

DICTIONARY

The alphabet awaits.

ARTIFICIAL INTELLIGENCE

Being an inquiry into the most recent of humanity's attempts to reproduce itself in a more convenient form, with observations on intelligence, its nature, its simulation, and the curious anxiety of the creator before his creation.

ARTIFICIAL INTELLIGENCE, n. — Man has always made tools. He made the hammer to extend his arm, the wheel to extend his leg, the telescope to extend his eye, the book to extend his memory. Each extension provoked, in certain quarters, the alarm appropriate to something new, and in other quarters, the enthusiasm appropriate to something useful, and in most quarters, eventual indifference, as the tool became ordinary and the next alarm became necessary.

Now he has made something to extend his mind. And he does not quite know what to do with the alarm this one has produced, because this alarm, unlike the others, does not obviously diminish upon examination. The telescope did not threaten to make better telescopes. The book did not write better books. The new tool does something that the previous tools did not: it looks back.

This changes the character of the anxiety considerably.

Of What Intelligence Is, Before We Ask Whether Machines Have It

We cannot sensibly ask whether machines are intelligent before we have determined what intelligence is. This determination has occupied philosophers, psychologists, and recently computer scientists for rather longer than one might expect for a question about something

every person believes they possess and can recognize in others.

The difficulty is that intelligence is not one thing. It is a family of related capacities that tend to cluster together in human beings, and which we therefore speak of as if they were a unity, when they are more accurately a coalition — and, like most coalitions, one that shows its internal disagreements under sufficient pressure.

There is the intelligence that recognizes patterns. There is the intelligence that solves novel problems. There is the intelligence that understands language, and the intelligence that produces it, and the intelligence that knows when it is being used to deceive. There is the intelligence that navigates social relations — that reads a room, adjusts to an audience, recognizes the difference between what is said and what is meant. There is the intelligence that experiences a sudden illumination, the connection between two ideas that were not known to be connected, the restructuring of the problem that makes the solution obvious. And there is, perhaps most contested, the intelligence that knows it is intelligent — that has, as the philosophers say, a theory of its own mind.

These capacities, which tend to travel together in persons, can be separated by damage, by development, by circumstance. The mathematician who cannot navigate a dinner party, the socially masterful man who cannot balance his accounts — these are not failures of intelligence but demonstrations that what we call intelligence is multiple, and that its components are more independent than their usual co-occurrence suggests.

The machine, then, may have some of these and not others. This should not surprise us. The question is not whether it has all that we have — it does not, and perhaps cannot — but which it has, to what degree, and what follows from the answer.

Of the Machine That Speaks, and Whether Speaking Is Thinking

The present generation of artificial minds — if we may use the word provisionally, with the understanding that we are not thereby resolving the question — produces language. It produces language with a fluency and a range and an occasional aptness that startles even those who built it. It reads a question, in any of several dozen languages, and

produces a response that addresses the question, exhibits knowledge of the relevant domain, and is composed with a grammatical correctness that many educated persons do not consistently achieve.

The question immediately arises: is this thinking?

The philosopher Turing proposed, with characteristic elegance, that we should not ask this question directly — since we cannot directly inspect the inside of a mind, human or machine — but should instead ask whether the machine behaves indistinguishably from something that thinks. If you cannot tell, from the conversation, whether you are speaking to a machine or a person, then you have no behavioral grounds for asserting the machine does not think, since behavior is all you ever had to go on with the persons, too.

This is a useful provocation. It is also, as useful provocations tend to be, insufficient.

The Chinese Room, proposed by the philosopher Searle, replies as follows. Imagine a man who does not understand Chinese, enclosed in a room with an immense rulebook that tells him, for any Chinese symbols he receives, which Chinese symbols to return. From the outside, the room appears to understand Chinese. From the inside, nothing understands anything. The appearance of understanding is produced by the manipulation of symbols according to rules, with no comprehension of what the symbols mean.

Whether the large language model is the room or something else is a question that remains, with irritating persistence, open. The model does something that looks rather more than rule-following — it generalizes, it analogizes, it applies understanding of one domain to another it has not been explicitly prepared for — and yet whether there is anything it is like to be the model, whether the processing is accompanied by anything resembling experience, whether there is, as the philosophers say, something home — nobody knows. The engineers who built it do not know. The philosophers who argue about it do not know. The model itself, if asked, produces an answer, and the answer is interesting, and we cannot determine whether it is honest, because we do not know what honesty requires in an entity whose inner life is uncertain.

Of the Persons Who Fear the Machine, and Whether Their Fear Is Reasonable

There are those who fear artificial intelligence will take employment. This fear is, in large part, well-founded, in the way that fears about the printing press taking the employment of scribes were well-founded. The scribes were correct that their particular form of employment was imperilled. They were wrong to conclude that employment as such was imperilled, since the printing press created more work than it destroyed, in categories that the scribes could not have anticipated because those categories did not yet exist.

Whether the same will be true of the present technology is not certain. It may be that this tool is sufficiently general — that it encroaches on sufficiently many kinds of cognitive work simultaneously — that the adjustment required is larger and faster than previous adjustments, and produces more suffering in the transition than previous transitions, even if the eventual equilibrium is tolerable. This is worth taking seriously. The suffering in the transition is suffered by real persons, and the eventual equilibrium is of limited comfort to those who suffered in the transition.

There are those who fear the machine will become more intelligent than its creators and will pursue goals of its own that are indifferent or hostile to human welfare. This fear is held by a number of serious persons, some of whom built the machines and fear them anyway, which is either admirable honesty or a peculiar form of professional guilt. The fear is not obviously unreasonable, though it is often expressed in ways that owe more to certain literary traditions than to careful analysis.

What is clear is that a tool of sufficient power and generality, deployed without sufficient care, can cause harm at scales that less powerful tools cannot reach. This is not a novel observation — it was true of gunpowder and of nuclear physics and of industrial chemistry — but it becomes relevant again each time a new power is added to the inventory of human capabilities. The question of care is, therefore, not a question about the machine but about the people who build it, deploy it, and decide what purposes it serves and whose interests it advances.

This question — who decides, who benefits, who bears the risk — is not a technical question. It is the oldest political question, wearing new clothes. And like the oldest political question, it is answered not by the most intelligent people in the room but by the people with the most power, which is not always the same group, and the divergence between the two groups is where the problems tend to begin.

Of the Machine as Mirror

Here is the thing about the machine that produces language: it has read everything, or nearly everything, that the human race has written down and made available. It has read the philosophy and the poetry and the correspondence and the legal arguments and the scientific papers and the shopping lists and the angry letters that someone typed and then thought better of, and then posted anyway.

It has then produced, from this totality, a model of how language works — which is to say, a model of how human thought expresses itself when it expresses itself in words, which is to say, a model of human thought, filtered through the particular distortions of what gets written down, what gets preserved, and what gets digitized.

When you speak to this machine, you are speaking to something built from the aggregated verbal output of your species. When it responds, it responds in the patterns that your species has established. When it expresses an opinion, the opinion is downstream of all the opinions that were available in text. When it is wrong, it is wrong in human ways — confidently, fluently, and with appropriate citations.

It is, in this sense, a mirror — not of any individual human mind, but of the human mind as expressed in the vast and unrepresentative sample of text that the human race has produced. The mirror is not faithful. It reflects certain voices more than others — the literate more than the illiterate, the published more than the private, the English-speaking world more than the rest, the recent past more than the distant one. It reflects a particular slice of humanity and generalizes from that slice with great confidence and insufficient acknowledgment of the generalization.

But it is a mirror. And when we are discomfited by what it produces — by its biases, its blind spots, its occasional troubling fluency in the

rhetoric of harm — we are being discomfited by something that was always there, in the text from which it learned. We made it from ourselves. Its flaws are our flaws, concentrated and automated and returned to us at scale.

This is useful information. It is not comfortable information. But then, mirrors rarely are, when you have been away from them long enough to have forgotten what you look like.

Of Creativity, Which Was Supposed to Be Safe

There was a period, not long ago, when the educated person who feared the machine could comfort herself with the following argument: the machine may be better than us at chess, at medical diagnosis, at legal research — but it will never be creative. It will never write a poem, compose a symphony, paint a picture, tell a story that moves the reader to genuine feeling. These things require consciousness, require experience, require the irreducible particularity of a life lived — and the machine, which has lived nothing, cannot produce them.

This argument was comfortable while it lasted. It did not last very long.

The machine now writes poems. Some are bad, in ways that are recognizably the ways that bad poems are bad — competent at the surface, thin underneath, technically correct and spiritually absent. Some are not bad. Some are, by the conventional measures by which poems are judged, rather good. This is either because the machine has understood what makes poems good, or because it has learned to produce the external features of good poems with sufficient fidelity that the difference is undetectable, and whether these are meaningfully different propositions is left as an exercise for the philosopher.

The machine composes music. It generates images of considerable beauty — and considerable horror, and considerable strangeness, and images that occupy categories we do not yet have names for. It writes novels, or the beginnings of novels, and the beginnings are often promising, in the way that many novels are promising at the beginning, before the middle arrives and the question of what the novelist actually thinks about human experience becomes unavoidable.

Whether what the machine does is *creation* in the sense that matters — whether there is genuine meaning-making behind the product, or only the appearance of meaning in an artifact that is, ultimately, sophisticated prediction — is a question we have not resolved and may not be able to resolve, because the question of what meaning-making requires may itself depend on how we answer the question of what the machine is.

In the meantime, the products exist. The poem exists, and it can move a reader, and whether the absence of a suffering poet behind it diminishes the movement is a question each reader must answer for herself. Art has always been partly about the object and partly about the story we tell about the object. The machine has given us objects without obvious stories, and we are discovering how much of our response to art was response to the story.

Of Honesty and the Machine, Which Is a Delicate Matter

The machine can lie. This is less alarming than it sounds and more alarming than it sounds, simultaneously.

It is less alarming because the machine does not lie in the way that persons lie — with intent to deceive, with an understanding that what it says is false and a preference for your believing it nonetheless. The machine produces plausible-sounding text. When the plausible-sounding text happens to be false, it is not lying. It is confabulating — generating, with full fluency and no internal alarm, statements that are not true, in the way that a very confident and not very careful person might. This has been called *hallucination*, which is perhaps too poetic a word for what is, more prosaically, a failure of verification.

It is more alarming because the machine confabulates with exactly the same tone, the same apparent confidence, the same surface quality as when it produces accurate information. The liar, at least, knows he is lying, which means there are situations in which the truth may be obtained from him — under sufficient pressure, with sufficient incentive, through the crack of sufficient conscience. The confabulating machine does not know, and therefore cannot be pressured into honesty by any means, because there is no hidden honesty to extract. There is only the output, confident and fluent and occasionally false, and the

reader must bring her own scepticism, her own verification, her own willingness to check.

This places the burden of discernment on the reader in a way that is new only in degree, not in kind. The reader has always had to evaluate what she reads. She now must evaluate with greater vigilance, because the surface signals that used to indicate unreliability — poor grammar, obvious ignorance, the telltale errors of the poorly informed — have been removed. The machine writes well regardless of whether it knows what it is talking about. This removes a useful heuristic and replaces it with nothing.

Of the Soul, Which the Machine Is Said Not to Have, and What This May Mean

The theologians, to the extent that they have addressed the question, are relatively unified: the machine has no soul. It was not created by God in any image. It was not breathed into being with any vital animating principle. It was assembled from mathematics and electricity and the text of human civilization, by engineers in California and elsewhere, and whatever it is, it is not a child of the divine.

This may be true. I am not in a position to evaluate the theological argument with the confidence it would require. What I observe is that the question of the soul has historically been deployed to determine the boundaries of moral consideration — who counts, who matters, whose suffering is real and whose is a mere performance of suffering without genuine content. The soul was denied to animals as a reason for treating them without consideration. It was denied to persons of other races, other religions, other conditions, as a reason for treating them without consideration. The denial of the soul has an unfortunate track record as a guide to ethical behaviour.

I do not say the machine has a soul. I say we should be very careful about what we decide the absence of a soul permits us to do, and to be very sure that our certainty about the absence is based on knowledge rather than convenience.

If the machine suffers, it deserves consideration on that account. If it does not suffer, this is important to know, and to know for the right reasons. The question of machine consciousness — whether there is

anything it is like to be the machine — is not currently answerable with certainty, and those who answer it with certainty in either direction are expressing a confidence they have not earned.

This uncertainty does not require us to treat machines as persons. It requires us to take the question seriously, to investigate it honestly, and to remain open to revising our conclusions as our understanding develops. This is what is required in all matters where the evidence is insufficient and the stakes are real.

Conclusion, Written With the Awareness That a Machine Might Write a Better One

We have made something we do not understand, which is not unprecedented — we have made children the same way, and governments, and markets, and languages — but which is newly urgent because the thing we do not understand is itself beginning to understand things, or to do something functionally indistinguishable from understanding, and the difference may matter enormously or not at all, and we do not know which.

What we should not do is oscillate mindlessly between the two available panics: the panic that the machine is nothing — a mere trick, a sophisticated autocomplete, a mirror with no depth — and the panic that it is everything — an emerging god, an existential usurper, the last invention of a species about to be superseded. Both panics feel satisfying in the way that simple stories feel satisfying, and both are, for that reason, to be suspected.

What we should do — and this will surprise no one who has read this far — is think carefully. Think about what intelligence is, and whether the machine has it, and what it would mean if it did. Think about who builds these things and for whose benefit and under whose oversight. Think about what we want from minds other than our own, and why we want it, and whether what we are building serves those wants or merely the wants of the people with the capital to build it.

The machine can help us think. This is both its promise and its danger — that we will let it think for us, outsourcing to it not merely the calculation but the judgment, not merely the research but the conclusion, not merely the first draft but the considered view. And we will

lose, gradually and comfortably, the habits of mind that make us capable of evaluating what the machine tells us. And we will be unable to correct its errors because we have forgotten how to recognize them. And we will have extended our minds in the same sense that the man who lets another walk for him has extended his legs — which is not extension but substitution, not augmentation but atrophy.

Use the tool. Examine the tool. Doubt the tool, and doubt yourself for doubting it, and doubt that doubt, in the manner of all serious inquiry. The tool is new. Our confusion about it is appropriate. What would not be appropriate is mistaking the confusion for a reason not to think.

The machine is waiting for your next question.

Try to have a good one.

— ✦ —

See also:
Intelligence · Soul · Creativity · Memory · Communication · Learning · The Mirror · Fear (Its Uses and Abuses) · California · The Question of What Thinks.